APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • LANDSLIDE

Location: Meppadi, Wayanad, Kerala, India • Generated: 06 Oct 2026 10:42 UTC • Status: APPROVED BY OPERATOR

Target Event	Landslide • Meppadi, Wayanad	**Risk Probability**	**86.9% (CRITICAL)**
Affected Area	16.63 km²	**Model Confidence**	95.0% Indicative
Exposed Population	37,734	**Antecedent Rainfall**	180 mm (72h: 572mm)
Terrain Slope	35° (Steep escarpment)	**SAM Scar Boundary**	Validated (7.03 km²)
Satellite Observation	Sentinel-1 SAR + Sentinel-2 MSI	**Human Verification**	APPROVED BY OPERATOR

Multi-Agent Intelligence & Evidence Progression

1. Satellite Ingestion: Copernicus Sentinel-1 C-SAR IW mode acquired pre- and post-event passes. Dense monsoon cloud cover (48%) partially degraded Sentinel-2 optical bands; radar penetration enabled uninterrupted surface change mapping.
2. DEM Terrain Modeling: Horn algorithm computed local slope exceeding 35° with converging concave profile curvature (-0.04), predisposing regolith to catastrophic mass wasting.
3. SAM Segmentation: Delineated spatial runout path of 7.03 km² encompassing Mundakkai and Chooralmala.
4. XGBoost & Bayesian Sequential Update: Initial baseline hazard belief (0.20) was updated sequentially via SAR decorrelation (+0.25), extreme 72h rainfall burst (+0.23), and terrain slope instability to yield a calibrated posterior risk of **91%**.

Response Recommendations & Relocation Planning

Designation	Capacity / Occupancy	Distance / Road Status	Hazard Exposure
Government School Meppadi (Primary)	650 / 380 (270 Available)	2.4 km • Open	LOW RISK
Community Relief Hall West	420 / 310 (110 Available)	4.1 km • Debris Clearance	MODERATE
Vythiri Camp North	800 / 220 (580 Available)	6.8 km • Clear	LOW RISK

Accountable Operator Review & Audit Trail

Reviewing Desk: APERIO Response Operations Desk
Operator Notes: Operator confirmed sensor convergence. Priority advisory drafted. Emergency services notified of bridge disruption.
Operational Boundary: Decision Support Output only. All evacuations require local ground confirmation.